

Internet and Networking for QA

DNS, IP, ports, HTTP/HTTPS, TLS, proxy, VPN and latency from a testing perspective.

From URL to application

When a URL is opened, the browser resolves the domain through DNS, connects to a server and exchanges requests/responses. Understanding the path helps separate application, network and infrastructure issues.

IP, ports and protocols

IP identifies network interfaces; ports help direct traffic to services; HTTP defines Web communication and HTTPS adds protection through TLS.

Proxy, VPN and firewall

These components can change routes, access and behavior. An issue that occurs only on a corporate network may not be an application defect.

How to investigate

Use DevTools Network, curl, DNS resolution and logs as appropriate. Capture time, environment, endpoint, status and latency when reporting connectivity issues.

When to use it

Use this knowledge when product risk, architecture or the delivery stage requires this type of validation. Confirm the goal, environment, required data and expected evidence before starting.

How QA applies it

Start from the behavior or risk to validate, prepare data and environment, execute the technique or tool reproducibly and capture evidence that helps the team understand the result.

QA tip

Before logging an application defect, check whether DNS, network, proxy, VPN or environment is the real cause.

Official resources

Site oficial:

https://developer.mozilla.org/en-US/docs/Learn_web_development/Howto/Web_mechanics/What_is_a_web_server